

A Composite Gel Polymer Electrolyte with High Performance Based on Poly(Vinylidene Fluoride) and Polyborate for Lithium Ion Batteries

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A composite membrane based on electrospun poly(vinylidene fluoride) (PVDF) and lithium polyvinyl alcohol oxalate borate (LiPVAOB) exhibiting high safety (self-extinguishing) and good mechanical property is prepared. The ionic conductivity of the as-prepared gel polymer electrolyte from this composite membrane saturated with 1 mol L⁻¹ LiPF₆ electrolyte at ambient temperature can be up to 0.26 mS cm⁻¹, higher than that of the corresponding well-used commercial separator (Celgard 2730), 0.21 mS cm⁻¹. Moreover, the lithium ion transference in the gel polymer electrolyte at room temperature is 0.58, twice as that in the commercial separator (0.27). Furthermore, the absorbed electrolyte solvent is difficult to evaporate at elevated temperature. Its electrochemical performance is evaluated by using LiFePO₄ cathode. The obtained results suggest that this gel-type composite membrane shows great possibilities for use in large-capacity lithium ion batteries that require high safety.

1. Introduction

Lithium ion batteries (LIBs) have gained an unprecedented importance in the last several decades as the energy storage of portable devices such as cell phones, laptop computers, etc.^[1,2] Furthermore, lithium ion batteries with large capacity are also promising power sources for hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEVs), pure electric vehicles (PEVs) and storage of wind, solar and tide energy in smart grids.^[3] However, large capacity battery modules require higher safety precaution due to the existence of organic liquid electrolytes, which are easy to get on fire leading to sometimes fires and explosions because of misuse, short circuits or local overheating.^[4] The quest for safer and more reliable electrolyte systems is therefore urgent, and solid electrolytes are promising candidates in this regard.^[5,6] If solid electrolytes can replace the organic electrolyte, safety will be greatly improved

in addition to other advantages such as no liquid leakage and long cycling life.

Solid electrolytes in lithium ion batteries mainly include inorganic Li⁺ ion conductors (ILICs) and polymer electrolytes (PEs). ILICs exhibit ionic conductivities in the range of 10⁻⁵ S cm⁻¹ to 10⁻¹ S cm⁻¹ at room temperature and are acceptable in microbatteries for applications such as microelectronics, medical implants, military industry, radio-frequency identification (RFID) and so on.^[3,7] In the case of large capacity battery modules, their processing performance is far from qualification due to poor mechanical strength and, sometimes, high sensitivity to humidity. Ever since the work of Wright and co-workers, who reported significant ionic conduction in a mixing system of alkali-halide with poly(ethylene oxide),^[8]

PEs (solid and gel types) have attracted a great deal of attention in terms of both fundamental and applied research. In the case of applications of solid PEs, there is still room for improvement because of low ionic conductivity, mostly ranging from 10⁻⁸ to 10⁻⁵ S cm⁻¹ at ambient temperature, and poor mechanical strength.^[9] Gel PEs (hereafter abbreviated as GPEs), which are formed by attracting a large amount of liquid electrolytes in polymer matrices and have characteristics of both the solid and the liquid electrolytes, have received increasing attention because of their good ionic conductivity (above 10⁻⁴ S cm⁻¹ at room temperature), wide electrochemical window, good thermal stability and good compatibility with both electrodes (anode and cathode) during cycling.^[10–12] Poly(vinylidene fluoride) (PVDF) and its copolymer poly(vinylidene fluoride-co-hexafluoropropylene) (P(VDF-co-HFP)), poly(ethyleneoxide) (PEO), polyacrylonitrile (PAN) and poly(methyl methacrylate) (PMMA) have been widely studied as polymer matrices for the GPEs in order to make them feasible for commercial applications.^[13–24] However, the application of most GPEs in large-scale systems is currently limited because of poor mechanical strength.^[25]

By virtue of its attractive properties, PVDF has been chosen as a polymer host for lithium ion battery applications.^[26] PVDF-based GPEs are very electrochemically stable due to the presence of a strong electron-withdrawing functional group (–C–F) and the high dielectric constant ($\epsilon = 8.4$), which helps for larger dissolution of lithium salts and subsequent support of high concentration of charge carriers. Many preparation methods

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and modifications were tried for PVDF and its copolymers. The ionic conductivities were generally high enough to envisage technological applications in battery systems.^[27–30] The properties of PVDF-based GPEs, such as porosity, tortuosity and uniformity of pore distribution, are strongly dependent on their processing methods. Compared to the GPEs from the methods such as solvent casting, plasticizer extraction and phase inversion, the GPEs from electrospun methods show higher porosity, higher electrolyte uptake and higher ionic conductivity due to the presence of fully interconnected pore structure.^[31] Electrospinning is a simple and versatile fabrication tool for preparing fibrous polymer membranes with fiber diameters ranging from several micrometers to tens nanometers. The GPEs based on electrospun PVDF membranes have shown potential applications in lithium ion batteries.^[32,33] However, the total cost should be decreased.

Lithium chelatoborates as electrolyte salts have been widely studied and with the aim of substituting LiPF_6 .^[34–37] The latter is unstable at elevated temperature and sensitive with a trace amount of water.^[38] Among the borates, lithium bis(oxalate) borate (LiBOB) and lithium oxalydifluoroborate (LiODFB) are promising candidates as electrolyte salts in lithium ion batteries.^[37,39] These lithium salts not only present merits such as low production cost, wide potential window and good thermal stability but also can form an effective solid electrolyte interface (SEI) film to stabilize the graphitic anode structure. Recently we reported a single-ion conducting polymer electrolyte, lithium polyvinyl alcohol oxalate borate (LiPVAOB, **Figure 1**), whose monomer is structurally similar to LiBOB.^[40] The PE presents single ion conductive behavior, but its ionic conductivity at ambient temperature is very low ($<10^{-5} \text{ S cm}^{-1}$ at room temperature) and mechanical strength is also poor. These drawbacks prevent it from practical applications in lithium ion batteries.

In this study, we used the single-ion conductor (LiPVAOB) and electrospun PVDF membranes to form a composite membrane, PVDF-LiPVAOB, whose structure is just like a commercial trilayer separator (polypropylene/polyethylene/polypropylene). The composite membrane has the characteristics of the electrospun PVDF (high electrolyte uptake) and the polyborate (high Li^+ ion transference number) and shows a very good mechanical properties. The safety is also much increased in comparison with the commercial separators. The ionic conductivity of the GPE based on the PVDF-LiPVAOB composite at room temperature is higher than that for traditional separator, Celgard 2730, saturated with LiPF_6 electrolyte. Moreover, the lithium ion transfer number in the prepared GPE at room temperature is markedly increased due to the existence of the single-ion conductor. The GPE also exhibits good electrochemical performance for LiFePO_4 . The composite membrane shows great possibilities for use in large-capacity lithium ion batteries.

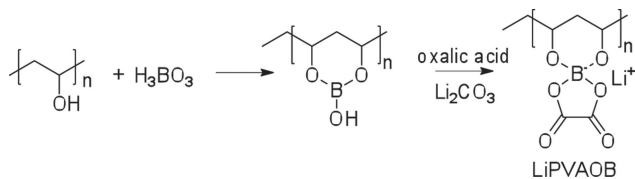


Figure 1. The preparation process for the polyborate LiPVAOB.

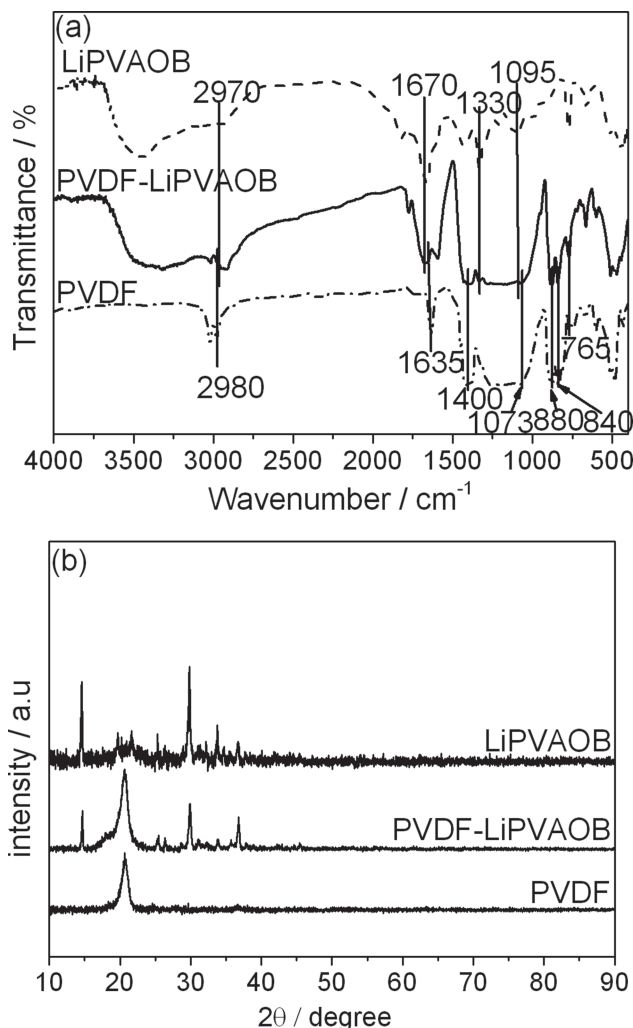


Figure 2. a) IR spectra and b) XRD of the electrospun PVDF, the prepared LiPVAOB and the PVDF-LiPVAOB composite membrane.

2. Results and Discussion

Figure 2a shows the IR spectra of the electrospun PVDF, the prepared LiPVAOB and the PVDF-LiPVAOB composite. The typical peaks of PVDF are situated at 2980, 1400, 1073, 880, 840 and 765 cm^{-1} and are attributed to $-\text{CH}_2-$ stretching, $-\text{CH}_2-$ deformation, C–C stretching, amorphous phase band, $-\text{CH}_2-$ rocking and $-\text{CF}_2-$ bending and skeletal bending, respectively.^[41,42] The peak at 1635 cm^{-1} can be assigned to the C=C absorption peak.^[42] The absorption peaks of LiPVAOB are situated at 2970, 1670, 1330 and 1095 cm^{-1} , attributing to C–H stretching, C=O stretching, B–O anti-symmetric stretching and $-\text{C}-\text{C}-\text{O}-$ stretching in the polymer chain, respectively.^[40] The peak of 3400 cm^{-1} is the $-\text{OH}$ stretching caused by the absorbed water in air due to the existence of Li^+ ions and its strong polarity. The appearance of all typical peaks of PVDF and LiPVAOB in the spectra of the PVDF-LiPVAOB composite indicates that the composite membrane is successfully prepared.

X-ray diffraction (XRD) patterns of electrospun PVDF, the prepared LiPVAOB and the PVDF-LiPVAOB composite are

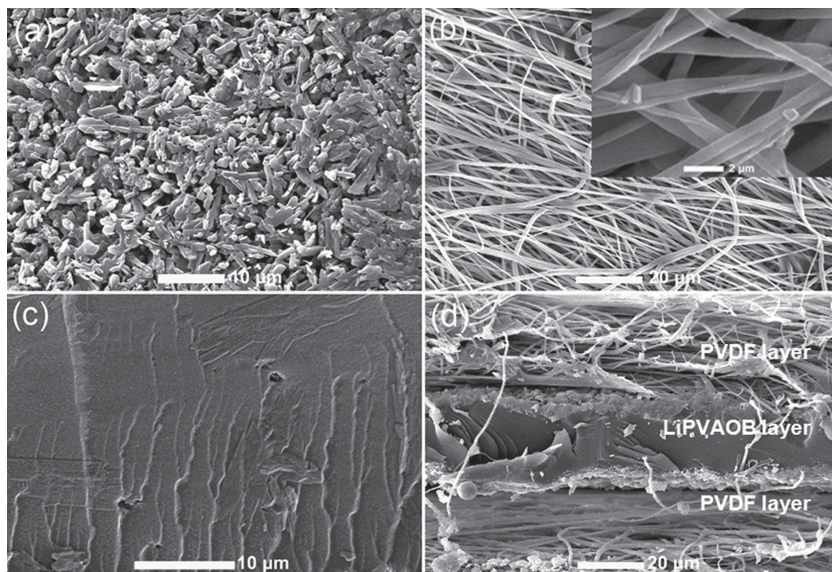


Figure 3. SEM images of a) the surface and c) the cross-section of the LiPVAOB membrane. SEM images of b) the surface and d) the cross-section of the PVDF-LiPVAOB composite membrane.

shown in Figure 2b. It is well known that the PVDF exhibits a semicrystalline structure with a strong characteristic diffraction peak at 20.8° , corresponding to the diffraction of (110) and (200) planes of its β -phase.^[43] The prepared LiPVAOB shows peaks at 14.5° , 18.5° – 23.5° , 25.4° , 26.3° , 30° and 36.5° , which are consistent with the previous reports.^[40] All the characteristic diffraction peaks also appear in the PVDF-LiPVAOB composite membrane. Of course, their crystallinity is not high.

Scanning electron microscopy (SEM) images of the prepared LiPVAOB and the PVDF-LiPVAOB composite membrane are shown in **Figure 3**. The surface of the LiPVAOB membrane (Figure 3a) is very rough. The inner structure of the membrane is uniform and compact from the cross-section (Figure 3c). The compact structure makes the membrane have low uptake ability (28.1 wt% for propylene carbonate).^[40] After forming the composite with the electrospun PVDF fibers, a porous PVDF was observed on the surface (Figure 3b). The typical pore size is below $2\ \mu\text{m}$ with a certain uniformity (the inset of Figure 3b). From the cross-section, a well formed trilayer or sandwiched structure can be seen and the thickness is about $70\ \mu\text{m}$. Because of the existence of the compact LiPVAOB layer between the PVDF layers, the particles of the electrode materials will not pass through the composite membrane while the Li^+ ion can be transported freely,^[40] which is of great importance to prevent possible micro-short-circuiting caused by the fine particles from the active electrode materials.

Thermogravimetry (TG) and differential scanning calorimetry (DSC) curves of the Celgard 2730, the electrospun PVDF, the prepared LiPVAOB and the PVDF-LiPVAOB composite membrane are shown in **Figure 4a,b**. The PVDF-LiPVAOB composite membrane is thermally stable up to 100°C , which is about 30°C lower than that of commercial separator, Celgard 2730. The main reason is that thermal stability of the composite is decided by the LiPVAOB component, which slightly decomposes when the temperature is up to 100°C , and the

main decomposition of the polymers happens at about 170°C ,^[40] the same with that of PVDF.^[28]

The TG and DSC curves of the PVDF-LiPVAOB GPE and the Celgard 2730 separator soaked with the same amount of the electrolyte ($1\ \text{mL g}^{-1}$) are shown in Figure 4c,d. The retention ability of the liquid electrolytes is crucial to the safety of lithium ion batteries. The organic electrolyte in the commercial separator starts to evaporate at 65°C , which is indicated by the sharp endothermic peak in the DSC curve of Figure 4d. In the TG curve it also presents a marked weight loss at the same temperature. When the temperature arrives at 112°C , the evaporable solvents disappear completely. The emission of a large amount of combustible gas at temperatures below 120°C is the main reason for many accidents including fires and explosion of lithium ion batteries.^[28] This also suggests that at temperatures above 120°C lithium ion batteries will not work due to very low ionic conductivity. The so-called shut-down

behavior of polyethylene in the commercial separator practically amount to any improvement for the safety of lithium ion batteries, which has also been well proven by a number of accidents, such as fires and explosions of lithium ion batteries. However, formerly there is no doubt about the real effect of the shut-down behavior of the polyethylene. In the case of the PVDF-LiPVAOB GPE, the absorbed electrolyte begins to evaporate at 85°C , which is 20°C higher than that of Celgard 2730. The main reason is that the organic electrolyte is absorbed and retained by the composite membrane. Instead, in the commercial separators, the liquid electrolyte is only dispersed in the pores. Moreover, the evaporation rate is much slower than that for the commercial separator with the same amount of liquid electrolytes. When the temperature increases to 140°C , the PVDF-LiPVAOB GPE shows only 35% weight loss, which is from the slight decomposition of LiPVAOB and evaporation of the liquid electrolytes. That is, a large proportion of electrolyte is still retained in the gel composite membrane even at 140°C . This result suggests that the lithium ion battery can still operate at 140°C . When the temperature is further increased, PVDF will melt down. This shut-down behavior can prevent the pass-through of the remaining liquid electrolytes, which will be of great value to improve the practical safety for lithium ion batteries.

The combustion test of the Celgard 2730 and the PVDF-LiPVAOB composite membrane is shown in **Figure 5**. The flame retarding ability of the membranes/separators is a critical factor for the safety of the lithium ion battery and has been very rarely mentioned because most separators or membranes are combustible. When the commercial separator of the Celgard 2730 was set on fire, the separator shrank immediately and was on fire within short time ($<3\ \text{s}$) (Figure 5c). This is similar to other traditional separators due to the existence of the combustible polyolefin matrix such as polyethylene or polypropylene. The composite membrane shows perfect flame retarding

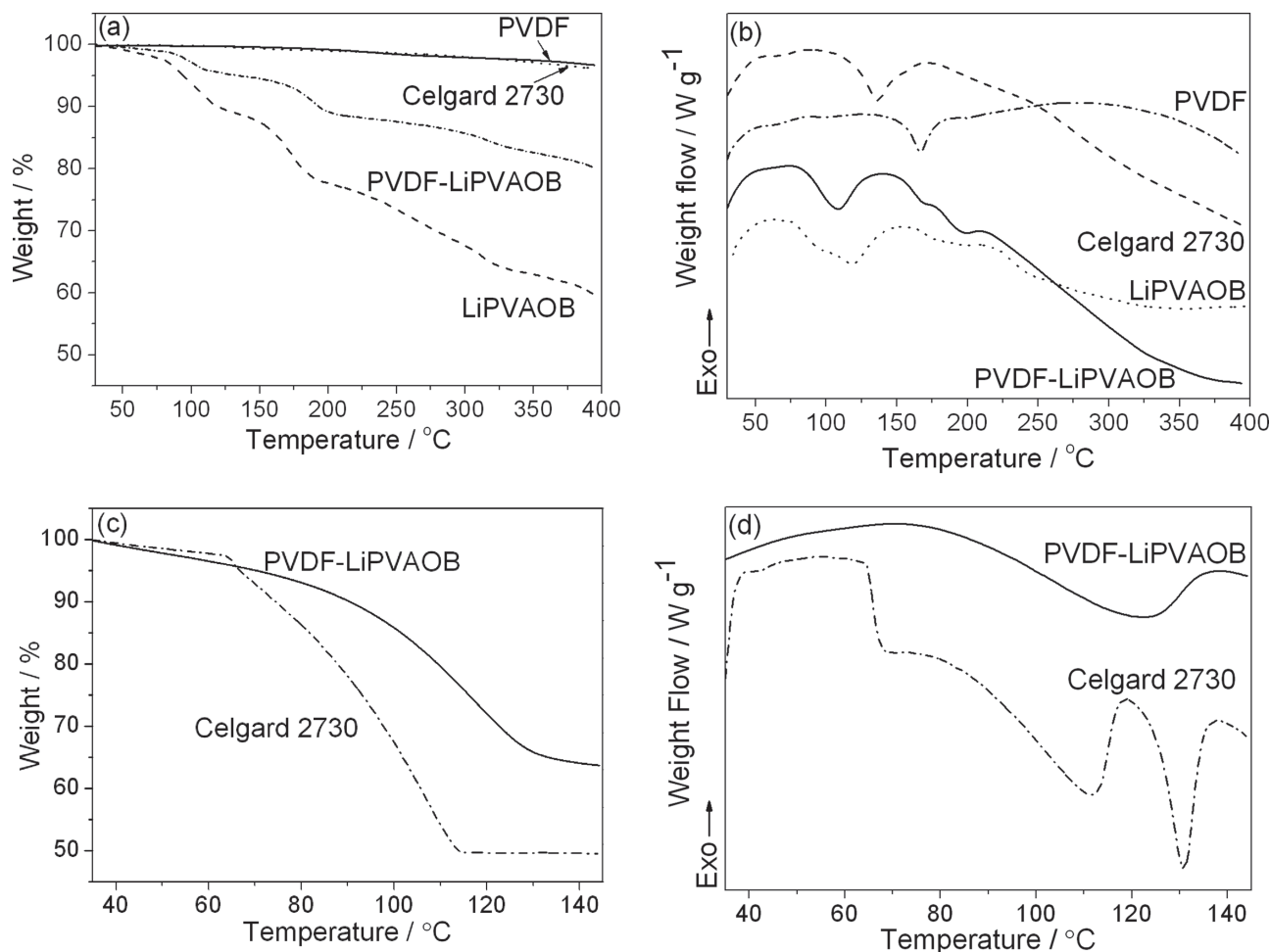


Figure 4. a) TG and b) DSC curves of the Celgard 2730, the electrospun PVDF, the prepared LiPVAOB and the PVDF-LiPVAOB composite membrane under air and the temperature rising rate is 10 °C min⁻¹. c) TG and d) DSC curves of the Celgard 2730 and the PVDF-LiPVAOB composite membrane (20 mg each) after absorbing the same amount of LiPF₆ electrolyte (20 μ L for 20 mg membrane) at the temperature rising rate of 2 °C min⁻¹ under nitrogen.

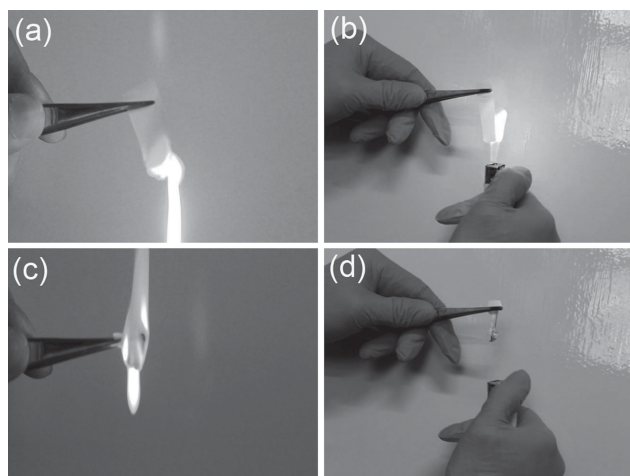


Figure 5. The combustion test of a,c) the Celgard 2730 and b,d) the PVDF-LiPVAOB composite membrane.

ability. It did not catch fire when set on fire (Figure 5d), although it also shrank. The main reason is that the PVDF cannot be easily set on fire and the LiPVAOB matrix is sandwiched by the PVDF layers. In the same time, the combustion test of the Celgard 2730, PVDF and the PVDF-LiPVAOB under the electrolyte-swollen state also were done in the air. When the membranes/separator (width: 1.5 cm and length: 10 cm) saturated with the electrolyte were set on fire, the absorbed liquid electrolyte was on fire immediately. The wet Celgard 2730 separator shrank immediately and was on fire in a very short time, similar to what occurred with the separator without electrolyte. The electrospun PVDF and PVDF-LiPVAOB membrane did not catch fire immediately when set on fire because the flame retarding ability of PVDF. The whole combustion times of the membranes/separator under the electrolyte-swollen state were 7.79 s (Celgard 2730), 14.24 s (PVDF) and 11.25 s (PVDF-LiPVAOB). This property also provides a higher safety performance for the composite membrane compared to the commercial polyethylene or polypropylene separators.

The stress-strain curves of the Celgard 2730, the wet Celgard 2730 (Celgard 2730 saturated with electrolyte), the electrospun PVDF, the wet electrospun PVDF (PVDF saturated with electrolyte), the prepared LiPVAOB, the PVDF-LiPVAOB composite membrane and the PVDF-LiPVAOB GPE are shown in Figure 6a. It is well-known that the poor mechanical strength limits the application of most GPEs, especially in large-scale energy storage systems. The maximal stress and strain of the PVDF-LiPVAOB GPE are 32.4 MPa and 3.1%, respectively. The stress value is very close to that for the wet Celgard 2730 (38.5 MPa), indicating that the PVDF-LiPVAOB composite membrane is acceptable for lithium ion batteries. In comparison with the electrospun PVDF, the strain of the composite is decreased. The main reason is that the LiPVAOB matrix between the PVDF layers has poor mechanical strength and limits the strain properties of the composite membrane. In addition, the membranes/separators have almost the same mechanical properties no matter what state (dry or wet) they are in. That is, the mechanical strength of the PVDF-LiPVAOB GPE is not influenced by the gelation. The main reason is that the existence of the electrospun PVDF in the composite membrane keeps the mechanical property of the gel PVDF-LiPVAOB. The gelation of PVDF with the electrolyte is due to the electrostatic interaction between the solvents and the surface polaric groups of $-C-F$, which is evidenced from the change of the above FT-IR. As a result, the polymer host is not changed much, leading to not much difference in the mechanical strength before and after the gelation. This is different from the gelation of polyacrylonitrile with the electrolyte, where the electrolyte can dissolve the polymer, leading to a sharp decrease in the mechanical strength.^[16]

The uptake amount of the liquid electrolyte for the composite membrane can be up to 88.5 wt%, which is near to that for the Celgard 2730 (90.9 wt%). This indicates that the ionic conductivity, which is mainly dependent on the amount of the organic electrolytes, will be at least on the same order of magnitude as the liquid electrolyte.

Figure 6b shows the dependence of the ion conductivities on temperature for the PVDF-LiPVAOB GPE, LiPVAOB and the commercial separator (Celgard 2730) saturated with the organic electrolyte at the temperature range from 25 to 75 °C. The ionic conductivity was calculated from the impedance plots shown in the insets of Figure 6b. The typical impedance plots consist of a high frequency semicircle followed by a low frequency straight line, which correspond to the bulk/grain boundary resistances and the diffusion of lithium ions, respectively. Here, the semicircular portion disappears indicating that the current carriers are ions and hence the total conductivity is the result of ionic conduction.^[44] The resistance of the bulk electrolyte has been retrieved from the intercept of the straight line on the real axis.^[45] The ionic conductivity of the LiPVAOB saturated with the $LiPF_6$ organic electrolyte at ambient temperature is $7.86 \times 10^{-5} \text{ S cm}^{-1}$, higher than that of the LiPVAOB saturated with the PC ($6.11 \times 10^{-6} \text{ S cm}^{-1}$).^[40] The ionic conductivity of the PVDF-LiPVAOB GPE at 25 °C is 0.26 mS cm^{-1} , a little higher than that of the Celgard 2730 saturated with the organic electrolyte (0.21 mS cm^{-1}). These data clearly present that the ionic conductivity of the GPEs are at the same level for the commercial separators, which is also satisfactory for practical lithium

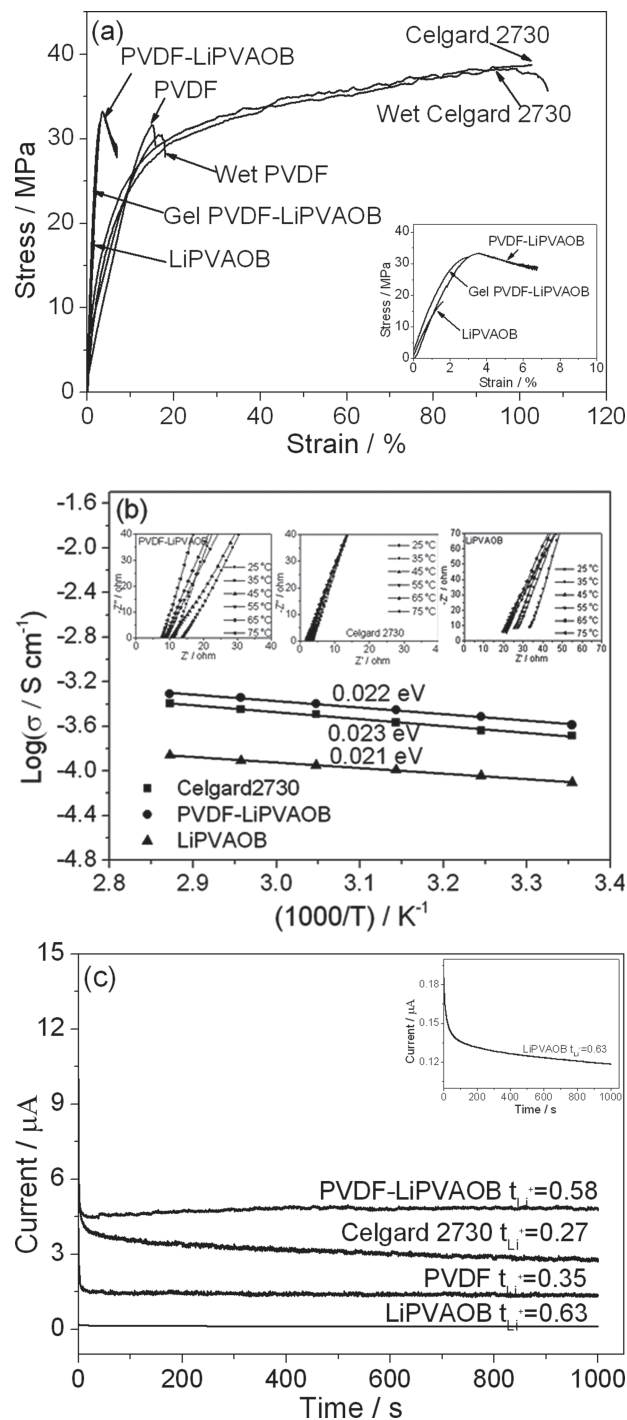


Figure 6. a) Stress-strain curves of the Celgard 2730, the wet Celgard 2730 (the Celgard 2730 saturated with the liquid electrolyte), the electrospun PVDF, the wet electrospun PVDF (PVDF saturated with the liquid electrolyte), the prepared LiPVAOB, the PVDF-LiPVAOB composite membrane and the PVDF-LiPVAOB GPE. b) Impedance plots of the conductivity data at different temperatures and Arrhenius plots of the PVDF-LiPVAOB GPE, LiPVAOB and the Celgard 2730 saturated with 1 M $LiPF_6$ organic electrolyte. c) Chronoamperometry profiles for the Celgard 2730, electrospun PVDF and LiPVAOB saturated with the liquid electrolyte and the PVDF-LiPVAOB GPE at 25 °C in block cells using Li metal as both electrodes with step potential of 10 mV.

ion batteries. The dependence of ionic conductivity on temperature can be reasonably fitted by the Arrhenius equation:

$$\sigma = A \exp(-E_a/RT) \quad (1)$$

where A is the pre-exponential factor and E_a is the activation energy. The calculated E_a values are 0.022 eV, 0.021 eV and 0.023 eV for the PVDF-LiPVAOB GPE, LiPVAOB and the Celgard 2730 saturated with the liquid electrolyte, respectively. That is, the movement of ions in the gel polymer chains needs almost the same energy as that in the organic electrolytes.

The Li^+ ion transference number estimated by chronoamperometry is shown in Figure 6c. The major drawbacks of dual ion conducting in lithium ion batteries are the lower lithium ion transference number and the higher electrodes polarization caused by the moving anions. In addition, the mobile anions take part in undesirable side reactions at the electrodes, which also can directly affect the performance of batteries. Therefore it is necessary to increase Li^+ ion transference number to gain good performance for lithium ion batteries. The calculated Li^+ ion transference numbers by comparing the initial and final current values are 0.27, 0.35, 0.63 and 0.58 for the Celgard 2730, the electrospun PVDF and LiPVAOB saturated with the LiPF_6 liquid electrolyte, and the PVDF-LiPVAOB GPE, respectively. The polarity of the PVDF matrix contributes to the improvement of the Li^+ ion transference number. That is, the Li^+ transference number of the PVDF (0.35) is slightly higher than that of the commercial separator (0.27). Because of the existence of single ion conductor in the composite GPE, the gel membrane can hinder a part of PF_6^- anions from passing through to give higher Li^+ ion transference number. As a result, the conduction contributed by Li^+ ions is 0.15 and 0.057 mS cm^{-1} for the PVDF-LiPVAOB GPE and the Celgard 2730 saturated with the liquid electrolyte. This means that the Li^+ ions in the GPE can move at the rate of about three times as those in the traditional separators. Formerly many researchers misunderstood that the movement of Li^+ ions in the more viscous gel media would be more difficult than that in the liquid electrolyte. Here it shows that it is not the case. As mentioned above, the single cation conduction of the LiPVAOB and the polar PVDF matrix greatly limit the movement of anions and promote that of Li^+ ions. In addition, the activation energies of the movement of ions in the PVDF-LiPVAOB GPE and the Celgard 2730 saturated with the liquid electrolyte are almost the same, 0.022 and 0.023 eV, respectively. However, the ions include anions and cations. It is clear that the anions are more difficult to move than the Li^+ ions in our PVDF-LiPVAOB GPE. Consequently, the activation energy for the movement of Li^+ ions will be less than that for the total ions. As to the definite method to detect it, much effort is needed.

In the case of the electrochemical stability of the GPE, it is similar to that of the Celgard 2730, about 4.8 V (Figure S1, Supporting Information), since it is mainly dependent on the liquid electrolytes. This wide electrochemical window is enough for lithium ion batteries.

The electrochemical performance of the PVDF-LiPVAOB GPE was evaluated by using LiFePO_4 as the cathode and Li metal as the counter and reference electrodes. Figure 7 shows the comparison of the cycling behavior for the PVDF-LiPVAOB

composite membrane and the Celgard 2730, electrospun PVDF saturated with the organic electrolyte. The corresponding charge-discharge curves for some selected cycle numbers are inserted into the figure. The reversible capacity of LiFePO_4 is about 120 mAh g^{-1} at 0.2 C for the GPE (Figure 7a), which is higher than that for the commercial separator (about 100 mAh g^{-1} , Figure 7b) and PVDF (about 95 mAh g^{-1} , Figure 7c). Evidently, this is due to the higher conduction of Li^+ ions, which makes the PVDF-LiPVAOB GPE show better capacity retention than that of the commercial separator. In addition, the discharge capacity in the first cycle for the Celgard 2730 saturated with the electrolyte is less than that in the successive cycles, while it does not happen in the GPE. The main reason is that LiFePO_4 using the Celgard 2730 as separator needs more time to make the liquid electrolyte to disperse uniformly when the Celgard 2730, PVDF and the PVDF-LiPVAOB membrane are added with almost the same amount of the liquid electrolyte. After several cycles, the discharge capacity of LiFePO_4 using the Celgard 2730 as the separator becomes stable. The cycling performance of the LiFePO_4 cathode using the GPE is very good, similar to that using the commercial separator. During cycling there is still no evident capacity fading.

From the corresponding charge-discharge curves, typical flat-shaped voltage profiles are observed at around 3.3–3.5 V in the case of the GPE (the inset of Figure 7a), which are consistent with the reported two phase coexistence reaction for the LiFePO_4 cathode and the difference between charge and discharge curves is very small, less than 0.2 V.^[46,47] In the case of the Celgard 2730, the voltage profiles of LiFePO_4 are also flat (the inset of Figure 7b). However, the difference between charge and discharge voltages is larger, at least 0.8 V. Evidently, this higher voltage difference is due to the polarization caused by lower transference number of Li^+ ions. In the traditional testing of the electrochemical performance of the LiFePO_4 cathode, the cathode is immersed in the liquid electrolytes and the reported voltage difference is similar to that for the GPE, less than 0.3 V at low current densities.^[46,47] The reason for the larger overpotential is that the LiFePO_4 electrodes used in the test are thicker (17 mg cm^{-2}) than most reported (<2 mg cm^{-2}).^[46,47] This also means that the GPE from the composite membrane is suitable for thicker electrodes. The above results show that the PVDF-LiPVAOB GPE provides more moveable Li^+ ions, consequently the polarization and the resulting voltage difference are much smaller. Capacity retention of the lithium ion battery based on the PVDF-LiPVAOB GPE during cycling is very stable (Figure S2, Supporting Information).

In addition, Figure 7d shows the comparison of the rate performance of LiFePO_4 between 0.1 C to 1 C measured with the PVDF-LiPVAOB GPE and the Celgard 2730 saturated with the organic electrolyte. When charged at 0.2 C and discharged at 0.1 C, 0.2 C, 0.5 C and 1 C and 0.1 C, the capacity retentions of LiFePO_4 tested with the PVDF-LiPVAOB are 129.5, 120.5, 110.3, and 86.7 mAh g^{-1} , respectively, which are higher than those for the corresponding Celgard 2730 (108.7, 100.5, 89.5, and 74.7 mAh g^{-1} for 0.1 C, 0.2 C, 0.5 C and 1.0 C, respectively). The composite gel membrane has an evident better rate performance.

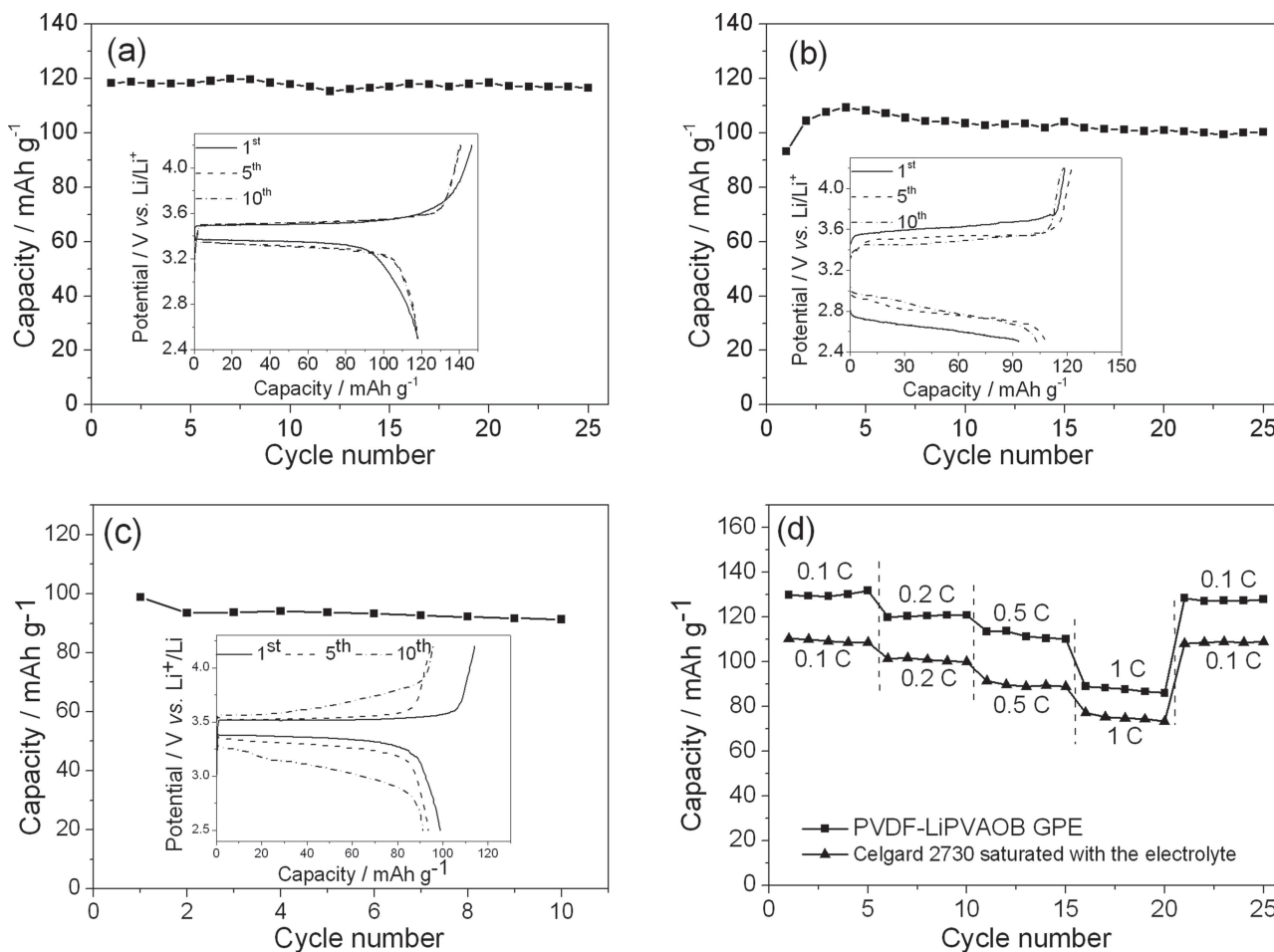


Figure 7. Electrochemical performance of LiFePO_4 tested by using a) the PVDF-LiPVAOB GPE, b) the Celgard 2730 and c) the electrospun PVDF as the separators, respectively, saturated with $1 \text{ mol L}^{-1} \text{ LiPF}_6$ electrolyte and Li metal as the counter electrode. d) The rate behavior of LiFePO_4 measured with the PVDF-LiPVAOB GPE and the Celgard 2730 saturated with the organic electrolyte at a charge current density of 0.2 C and discharge current density of 0.1 C, 0.2 C, 0.5 C, 1 C and 0.1 C, respectively.

3. Conclusions

A sandwiched composite membrane consisting of a poly(vinylidene fluoride) (PVDF) and lithium polyvinyl alcohol oxalate borate (LiPVAOB) have been prepared. The composite membrane exhibits good mechanical properties. It is gelled by liquid electrolyte to achieve gel polymer electrolyte (GPE). The ionic conductivity for Li^+ ions in the GPE at ambient temperature is two times higher than that in the commercial separator (Celgard 2730) saturated with liquid electrolyte, which is mainly due to the higher lithium ion transference number for the GPE. The composite matrix is difficult to set on fire and the liquid electrolyte in the GPE is also difficult to evaporate, which greatly improve the safety of lithium ion batteries. The GPE is evaluated using LiFePO_4 as the cathode and Li metal as the counter and reference electrode. The cathode delivers a higher discharge capacity at room temperature together with lower overpotentials in comparison with the commercial separator and good capacity retention. This PVDF-LiPVAOB GPE shows great possibilities for applications in large-scale lithium ion batteries requiring high safety and low cost.

4. Experimental Section

Synthesis of LiPVAOB: The procedure of preparation was based on the literature.^[40] According to the chemical equation in Figure 1, PVA (Sinopharm, degree of alcoholysis $\geq 98\%$, Mw: 105 000, 200 mg) and boric acid (Sinopharm, 148 mg) were dissolved into DMSO (Sinopharm, 20 mL) and heated to 80°C for 8 h to give a homogeneous and transparent solution. Then lithium carbonate (Sinopharm, 88 mg) and oxalic acid (Sinopharm, 300 mg) were added. The mixture was stirred at 100°C for 24 h. Cooled to room temperature, the solution was later cast on a flat glass plate at 70°C to remove the solvent to get membranes with a thickness of about $40 \mu\text{m}$.

Preparation of the PVDF-LiPVAOB Gel Polymer Electrolyte (GPE): The electrospun PVDF ($20 \mu\text{m}$ thickness, Yingke Separators Co., Ltd, Changzhou) was fixed on a glass plate and wetted with ethanol water solution (1/1, v/v). Then the LiPVAOB membrane was put above and wetted by ethanol water solution (1/1, v/v). Another electrospun PVDF layer was further put on the LiPVAOB membrane. The PVDF-LiPVAOB composite membrane (thickness: $70 \mu\text{m}$) was achieved after drying at 80°C for 5 h. It was punched into circular pieces ($d = 19 \text{ mm}$) and then dried under vacuum at 80°C for 48 h. After that, the pieces were soaked in an organic liquid electrolyte ($1 \text{ mol L}^{-1} \text{ LiPF}_6$ solution in EC/DMC/EMC (ethylene carbonate/dimethyl carbonate/ethyl methyl carbonate), 1/1/1, w/w/w, Zhangjiagang Guotai-Huarong New Chemical Materials

Co., Ltd) over 12 h in a glove box (water content < 1 ppm) to obtain the GPEs for further measurement.

Celgard 2730 (polyethylene, Celgard, thickness: 20 μm , pore size: 0.03–0.1 μm , porosity: 43%) was used as one control to evaluate the properties of the PVDF-LiPVAOB composite membrane.

Unless other stated, the following measurements were performed at room temperature. The diffuse reflectance infrared spectrometry (DRIFT) method was employed to measure infrared spectra (IR) of the samples. Before the IR measurements, the samples were dried at 80 °C under vacuum for 48 h again and then stored in a dry box for 2 h. The IR spectra (4000–400 cm^{-1} , resolution 2 cm^{-1}) were recorded with a BRUKER VECTOR-22 spectrometer. XRD was performed on a Bruker D8 X-ray diffractometer (50 KV, 50 mA) with a copper target ($\lambda = 1.5405 \text{ \AA}$) and Ni filter at the scan rate of 5° min^{-1} from 10° to 90°. The surface morphology of the prepared membranes was investigated by Philips XL30 SEM. The membranes were dipped into liquid nitrogen and broken into two parts, and then the SEM images of the cross-sections were recorded. All of the samples were sputtered with gold prior to the SEM measurement. TGA and DSC of the membranes including the commercial Celgard 2730 separator were carried out by utilizing a Perkin-Elmer TGA7/DSC7. The thickness of the membranes was measured with a micrometer (SM & CTW, Shanghai). Stress-strain test was conducted using a Sansi YG832 tensile testing machine with a crosshead speed of 1 mm min^{-1} . The width of the sample was 4 mm.

The amount of liquid electrolyte uptake (η) is calculated as the following Equation (2):

$$\eta = (W_t - W_0) / W_0 \times 100\% \quad (2)$$

where W_0 and W_t are the weights of the membranes including the commercial Celgard 2730 separator before and after the absorption of the organic liquid electrolyte, respectively. The weight was weighed in a glove box.

The ionic conductivities were measured by electrochemical impedance spectroscopy (EIS). The samples were measured in blocking-type cells where the GPE membranes or the commercial Celgard 2730 separator saturated with the liquid electrolyte were sandwiched between two stainless steel electrodes. Impedance data were obtained with an electrochemical working station CHI660C (Chenhua) in the frequency range 10 Hz to 100 kHz between 25 and 75 °C. The ionic conductivity was calculated from the Equation (3):

$$\sigma = l / (R_b A) [\text{in S cm}^{-1}] \quad (3)$$

where σ is the ionic conductivity, R_b is the bulk resistance, l is the thickness of the GPE or the separator, and A is the area of the stainless steel electrode since the area of the GPE or the separator is larger than that of the steel electrode.

The linear sweep voltammogram was obtained by the electrochemical working station using a two-electrode cell. Stainless steel was used as the working electrode and lithium foil as the counter and reference electrode, respectively. The measurement was done between 0 and 6 V (vs. Li^+/Li) at the scan rate of 2 mV s^{-1} .

The chronoamperometry profile was obtained by the electrochemical working station measuring in blocking-type cells where the GPE or the Celgard 2730 separator saturated with the liquid electrolyte was sandwiched between two lithium metal electrodes. The step potential was 10 mV.

Charge/discharge and cycling tests of the LiFePO_4 cathode using the PVDF-LiPVAOB composite membrane, the Celgard 2730 and the electrospun PVDF as the separators, which are saturated with the liquid electrolyte, were conducted by assembling 2025 coin-type cell with lithium metal foil as the counter and reference electrode. The used cycler is a Land tester (CT2001A) and the voltage is between 2.5 and 4.2 V at a current density of 0.2 C based on the LiFePO_4 working electrode. The working electrode was prepared by coating the *N*-methyl-1-pyrrolidone (NMP)-based slurry containing LiFePO_4 (STL Energy Technology Co., Ltd.), acetylene black and PVDF (binder) in a weight ratio of 8:1:1 on an aluminium foil (thickness: 20 μm) using a doctor-blade technique, and

the cast foils were then punched into circular pieces ($d = 15 \text{ mm}$) and dried at 120 °C for 12 h under vacuum. The mass loading of LiFePO_4 was about 17 mg cm^{-2} , much higher than that for the most reported, about 2 mg cm^{-2} . The GPE or the Celgard 2730 saturated with the liquid electrolyte was used as the separator and the electrolyte. All cells were assembled in an Ar-filled glove box.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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